

Superclassical Thermodynamics

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1 Introduction

For nearly a century, orthodox quantum mechanics (often associated with the Copenhagen interpretation) has been universally accepted as the fundamental framework of microscopic physics. While its mathematical predictions are flawlessly accurate, its philosophical implications—such as intrinsic randomness, wave-function collapse upon observation, and the existence of abstract, multi-dimensional configuration spaces—have left physicists grappling with profound ontological paradoxes.

This document serves as a comprehensive introduction to an alternative, deeply realist paradigm known as **Superclassical Thermodynamics** (or Emergent Quantum Mechanics), primarily developed by Gerhard Grössing and his team . Originating from the principles of non-equilibrium thermodynamics and stochastic electrodynamics, this theory proposes a radical shift in perspective: quantum mechanics is not an ultimate, fundamental theory of nature, but rather an emergent macroscopic approximation of a deeper, chaotic, and strictly deterministic sub-quantum reality.

Inspired by macroscopic fluid-dynamics experiments—specifically, bouncing droplets on a vibrating fluid bath that mimic wave-particle duality—this theory replaces the abstract mathematics of orthodox quantum mechanics with tangible physical processes in real three-dimensional space . In this framework, the universe is permeated by a continuous zero-point energy field, and "particles" are not dimensionless points, but open, dissipative thermodynamic systems constantly exchanging heat with this fluctuating sub-quantum medium.

The Goal of this Document is to distill the complex mathematical and physical concepts of Superclassical Thermodynamics into an accessible, logically structured overview. It aims to demonstrate how the most baffling quantum phenomena—such as energy quantization, multiple steady states, non-locality, and the Heisenberg Uncertainty Principle—can be elegantly derived using purely classical physics and fluid dynamics.

Why is this Relevant? This theory is highly relevant to the foundational physics community because it successfully resolves the wave-particle paradox and the measurement problem without invoking new, mystical forces. By demonstrating that quantum mechanics can be understood as a classical diffusion process driven by vacuum fluctuations, it provides a deterministic ontology that restores a 3D physical reality, while actively acknowledging and addressing its own current limitations, such as its tension with Special Relativity

2 A theoretical Model of Superclassical Thermodynamics

Objects of the Theory: Particles and the Thermal Scalar Field

In subquantum thermodynamics, the traditional view of a point-like, inert, and isolated particle is discarded. It follows Bohm’s interpretation, but with a better explanation of how this pilot wave arises. The particle is modeled as an open and dissipative system that is maintained in a non-equilibrium steady state. Inspired by macroscopic experiments with bouncing fluid droplets, the physical particle is conceived as a “bouncer” that oscillates continuously with an intrinsic characteristic frequency.

The environment in which the particle is immersed is the thermal bath, which is constituted by the zero-point fluctuation energy field of the vacuum. This medium acts as an immense thermostat or thermal reservoir that permeates all space. The vacuum and the particle form an inseparable physical system: the thermal bath provides the constant energy flow that the particle needs to maintain its oscillation against friction, while the particle returns heat to the medium in a continuous exchange.

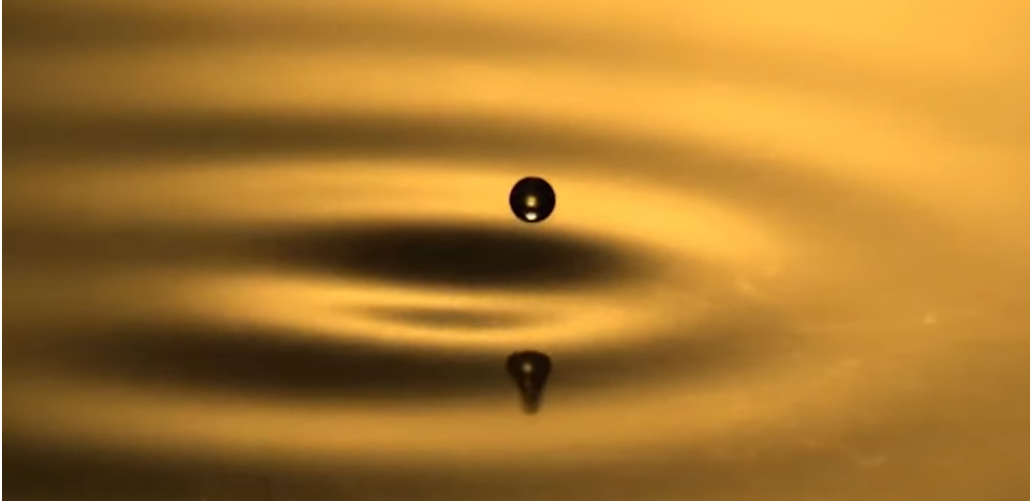


Figure 1: Analogy of the Bouncer

2.1 Particle Dynamics: Composition of Velocities

The dynamics dictating the particle’s motion are not governed by a single classical velocity vector, but by the coexistence of two independent velocity fields that are orthogonal on average. The emergent total motion is decomposed into:

- **Convective velocity (v):** Represents the macroscopic classical or forward motion. Mathematically, it is defined from the gradient of the particle’s action divided by its mass:

$$v = \frac{\nabla S}{m}$$

- **Diffusive or osmotic velocity (u):** Represents the rapid transverse fluctuations caused by the heat exchange with the vacuum. Its equation obeys a diffusion process and depends directly on the gradient of the probability density P (which maps the thermodynamic distribution):

$$u = -\frac{\hbar}{2m} \frac{\nabla P}{P}$$

A fundamental mathematical and physical rule of the model is the statistical orthogonality between both velocities ($v \cdot u = 0$). This property is crucial because it cancels the terms that would generate linear dispersion over time. Thanks to this orthogonality, a thermodynamic process that is stochastic, erratic, and irreversible at the microscopic level emerges at the macroscopic level as a reversible and ordered ballistic diffusion, making the system identical to the dynamics described by Bohmian mechanics and therefore the Schrödinger equation. As a result, this dynamics are compatible with the quantum experiments and expect the same results.

2.2 Dynamics of the Thermal Scalar Field and Mutual Interaction

The thermal scalar field (thermal bath) is not a passive container, but a dynamic physical medium whose evolution can be understood exactly as the superposition of two physical phenomena operating at different scales:

- **The stochastic background noise (Microscopic level):** At the shortest time scale, the vacuum exhibits random energy fluctuations (a “white noise” or thermal agitation). These stochastic impacts act as small uncontrollable forces that force the particle to perform a continuous erratic motion, completely analogous to Brownian motion. Because of this, the particle adopts the property of a “walker” moving through its environment.
- **The thermal diffusion waves (Emergent macroscopic level):** Simultaneously, every time the particle dissipates energy to maintain its steady state, it “heats” the surrounding medium locally. By averaging the collective effect of these millions of bounces

and microscopic interactions, the background noise self-organizes, emerging an ordered pattern of regular heat diffusion waves. These waves are directly modulated by the geometry of the environment (boundary conditions).

The wave-particle duality is resolved naturally and without paradoxes through a “symbiotic” relationship or relational causality. There is no abstract immaterial wave and a separate spherical particle. The background noise agitates the particle causing its Brownian motion; simultaneously, the energy that the particle dissipates structures this noise, creating a pattern of macroscopic diffusion waves in the vacuum (the “path excitation field”). At the end of the cycle, the particle’s trajectory is dynamically guided (“surfing”) on the very stationary diffusion waves that it and its past interactions have helped to sculpt in the medium.

3 Quantization

3.1 Continuous Energy Exchange

In the orthodox interpretation of quantum mechanics, the quantization of energy is introduced as an axiomatic mathematical postulate. In contrast, the superclassical model derives the Planck-Einstein relation ($E = \hbar\omega$) from purely classical non-equilibrium thermodynamics.

At the sub-quantum level, a particle is not a closed object possessing a static, discrete block of energy. Instead, it is an open, dissipative system (a “bouncer”) that maintains a non-equilibrium steady-state. To sustain its characteristic oscillation frequency (ω_0) against the “friction” of its environment, the particle must continuously absorb kinetic energy from the thermal bath of the zero-point field and constantly dissipate heat back into it.

The crucial insight is that this underlying exchange of energy with the medium is entirely continuous. However, because the particle is in a steady-state local equilibrium, the laws of thermodynamics dictate a strict energy balance over time. When you average this continuous, chaotic absorption and emission of heat over one full oscillation cycle, the total energy throughput always emerges exactly as $\hbar\omega_0$. The authors refer to this phenomenon as a “fleeting constancy”. Therefore, the discrete quantum of energy ($E = \hbar\omega$) is not a fundamental building block of nature, but rather the macroscopic average of a continuous thermodynamic process required to keep the particle oscillating.

3.2 Path Memory and the Quantization of Orbits

This continuous thermodynamic mechanism naturally extends to explain why a particle moving in a potential (such as an electron in an atom or a harmonic oscillator) can only occupy certain discrete spatial orbits.

When the “walker” moves along a closed trajectory, it continuously emits diffusion waves into the zero-point field. These waves do not instantly vanish; they superpose and create a physical “path memory” in the vacuum. As the particle orbits, it constantly interacts with the wave patterns it left behind in its own past.

If the particle were to move arbitrarily, the interference with its own path memory would disrupt its steady-state and destabilize the trajectory. Therefore, to maintain its energetic balance, the particle is forced to “self-trap” in specific, resonant orbits. A stable orbit is only possible if the particle undergoes an exact integer number (n) of additional periodic heat

exchanges (absorptions and emissions) during a single orbital period.

When the classical action of these continuous, adiabatic heat exchanges is integrated over a full cycle, it results in the discrete relation $\oint dS = nh$. This restricts the particle's available energy states to a discrete spectrum—such as $E(n) = (n + 1/2)\hbar\omega_0$ for the harmonic oscillator. Ultimately, quantized orbits are just the stable, resonant macroscopic patterns that arise from the continuous, classical heat exchange between the particle and its own wave wake.

4 Multiple Steady States

In subquantum thermodynamics, “multiple steady states” (which in orthodox quantum mechanics correspond to discrete or quantized energy levels) are not postulated as a mathematical axiom; instead, they emerge from the physical dynamics of an oscillator interacting with its thermal environment.

Below is a detailed explanation of this phenomenon, its mechanism through a practical example, and the limits of validity predicted by the theory.

4.1 What are Multiple Steady States?

In classical physics, a confined particle (such as an electron around a nucleus) could, in principle, possess any continuous amount of energy and orbit at any distance. However, in the superclassical model, the particle (the “bouncer”) is not an inert object but an **open dissipative system** maintained in a non-equilibrium steady state. To survive the “friction” of its environment and avoid coming to a halt, it requires a constant flow of energy: it absorbs heat from the fluctuations of the zero-point field and dissipates it back into the vacuum.

For this state to remain stable over time (steady), the laws of thermodynamics demand a perfect balance: the average heat exchange over a full oscillation cycle must be exactly zero. Multiple steady states are, therefore, the only geometric and energetic configurations (**resonant orbits**) in which this strict thermodynamic balance can be maintained.

4.2 The Physical Mechanism: Path Memory and Self-trapping (The Electron Example)

To understand how a particle ends up exclusively in one of these steady states, consider the example of an electron attempting to orbit an atom:

- **Path Memory:** As the electron orbits, its continuous bounces emit thermal diffusion waves into the vacuum field. These waves do not vanish instantly; they superpose and create a physical “path memory” in space. The electron constantly interacts with the wake of waves left in its own past.
- **Instability and Decay:** If the electron attempts to follow an arbitrary, continuous (non-resonant) orbit, its current motion will collide destructively with the wave pattern of its

path memory. This thermal interference breaks the delicate energy balance: the particle begins to dissipate more heat than it can absorb. Much like a satellite experiencing atmospheric friction, the orbit becomes thermodynamically unstable, loses kinetic energy (momentum), and “decays.”

- **Self-trapping:** This decay continues until the electron falls into a trajectory where its internal oscillations synchronize perfectly with the standing waves of the environment. In these stable orbits, the electron undergoes a strictly **integer number** (n) of additional periodic heat exchanges (absorptions and emissions) during a single orbital period. By doing so, the electron “self-traps” into a discrete orbit, and its energy stabilizes at an exact average value, such as:

$$E(n) = \left(n + \frac{1}{2}\right) \hbar\omega_0$$

Other orbits, being unstable, simply decay before they can be measured.

4.3 Where do Quantum States remain valid?

One of the most profound implications of this model is that it predicts exactly where and when quantum mechanics itself (including the Schrödinger Equation, the Born Rule, and the superposition principle) ceases to function.

The perfection of quantized orbits, the reversibility of the Schrödinger equation, and the statistical orthogonality between classical (v) and diffusive (u) velocities are **emergent properties** that only exist as long-term averages.

- **Macroscopic / Emergent Quantum Scale** ($t \gg 1/\omega$): When we observe the system at time scales much larger than the extremely fast oscillation period of the “bouncer” (for an electron, $\omega \approx 10^{21}$ Hz), the billions of stochastic and thermal impacts average out perfectly. At this scale, we see stable steady states, the Born rule is exactly satisfied, and wave behavior is smooth and ordered.
- **Subquantum Scale (The Collapse of Quantum Mechanics)** ($t \simeq 1/\omega$): The theory predicts that if we could increase our temporal resolution to observe the system in minuscule intervals comparable to a single individual “bounce” of the particle, the laws of quantum mechanics would collapse. At this time scale, the symmetry between diffusion currents breaks, and the statistical average ensuring orthogonality fails. Motion is no

longer dictated by a reversible wave function but is revealed as a **chaotic, stochastic, and purely irreversible thermodynamic process**.

In conclusion, the theory not only explains why multiple steady states exist based on thermodynamic forces and classical resonance, but asserts that quantum mechanics is solely a **mean field theory**. In the depths of the subquantum scale, the world ceases to be quantum and returns to the domain of a classical thermodynamics of fluctuating fluids.

However, we can't possibly test this because we would need too much resolution.

5 Entanglement and Non-Locality

In the superclassical model, entanglement is not a mysterious action at a distance, but a direct consequence of the physical existence of a nonlocal “path excitation field”. When analyzing multiple pathways or multiple particles, the theory utilizes the concept of “vacuum landscaping”. Setting up or altering an experimental arrangement changes the boundary conditions of the zero-point vacuum field, which instantly shapes the standing waves of the thermal landscape across the entire domain.

This can be done instantly because of the heat diffusion equation. This is a classical equation that exhibit a non-local behaviour.

Because the particles are constantly interacting with this shared sub-quantum bath, they are connected by an “entangling current” (J_e). This current is characterized by the difference in their diffusive velocities ($u_1 - u_2$) and drives the non-local correlations observed in experiments. Bell’s theorem dictates that any deterministic hidden-variable theory must be non-local to reproduce quantum mechanics. By explicitly introducing this overarching, interconnected thermal field that updates systemically, the theory openly embraces a deterministic, non-local ontology, thus perfectly complying with Bell’s inequalities.

5.1 The Problem of the Classical Diffusion Equation

While the model successfully reproduces quantum interference and entanglement, it relies heavily on the classical diffusion (or heat) equation. However, we know that this equation is just a mathematical model, that in the classical framework fails to explain localism. The authors openly acknowledge that this equation is an imperfect, idealized mathematical model because it yields the “physical artefact of infinite speed of field propagation”.

In classical physics, we know that true interactions are local and respect a speed limit. But the mathematical formulation of parabolic diffusion equations produces no traveling waves or phase velocity; instead, the entire physical domain “breathes” in phase with the oscillating source instantly. The authors use this classical artifact as a convenient tool to mimic quantum non-locality, creating a strict, idealized instantaneous connectivity that does not truly exist in standard classical mechanics.

However, there are still some powerful questions. Why can we say now that reality behave this way and not before in the classical world? What has changed?

5.2 The Conflict with Relativity and Bohmian Mechanics

This idealized non-locality leads to a significant foundational problem, which this theory shares with the de Broglie-Bohm interpretation. Both theories are based on an ontologically deterministic framework combined with non-local interactions.

The problem is that if a physical reality is both deterministic and non-local, it implies the existence of superluminal signaling. Even if we as macroscopic observers cannot exploit this to send messages faster than light—due to “epistemic indeterminism” (our inability to control the chaotic, exact microscopic thermal fluctuations)—the theory implies that “hidden” superluminal signaling is occurring on the sub-quantum level. This deterministic superluminal influence fundamentally clashes with Einstein’s Special Relativity, which prohibits any causal transmission of information faster than light.

5.3 The Proposed Solution: $1/\omega$ Time and Emergent Relativity

To solve the paradox between superluminal non-locality and relativity, the authors abandon the “idealized” infinite speed of the diffusion equation in favor of a more realistic “systemic nonlocality”.

They propose that when boundary conditions change (like Alice altering her interferometer), the vacuum landscape does not update strictly instantaneously. Instead, the standing waves of the zero-point field take a minute, non-zero amount of time to reconfigure, on the order of $t \sim 1/\omega$ (where ω is the particle’s extremely fast oscillation frequency).

Although this reconfiguration time is incredibly small, it still implies propagation velocities vastly exceeding the speed of light ($v \gg c$). To reconcile this with standard physics, the authors suggest a radical paradigm shift: just as orthodox quantum mechanics is proposed to be merely an emergent approximation of sub-quantum thermodynamics, standard Special Relativity might also be an emergent, macroscopic approximation of a deeper theory.

They argue that our current application of relativistic concepts might be too restrictive and idealistic at the sub-quantum scale. Consequently, they believe physics must develop an “emergent relativity”. By looking into frameworks like “double special relativity” or “superstatistics” (where physical laws change depending on the space-time scale), they hope to formulate a new relativistic theory where superluminal sub-quantum diffusion wave fields can exist without generating macroscopic causal paradoxes.

However, we know that for the moment this has not been done yet.

6 The Heisenberg Uncertainty Principle in Superclassical Thermodynamics

To achieve a perfect understanding of how the Heisenberg Uncertainty Principle emerges in the superclassical model, we must fundamentally dismantle the orthodox quantum mechanical explanation and replace it with a purely thermodynamic and statistical framework.

6.1 The Ontological Difference: Fuzziness vs. Hidden Determinism

In orthodox quantum mechanics, the uncertainty principle is taught as a direct consequence of the wave-particle duality and Fourier analysis: a particle is a wave, and mathematically, a wave cannot have both a perfectly defined position and a perfectly defined wavelength (momentum) at the same time. It represents an ontological indeterminism—the particle literally lacks a precise position and momentum.

The superclassical theory shatters this view. In this model, the particle is a localized, real physical entity (a “bouncer”) oscillating and moving through a real sub-quantum fluid (the zero-point field). At any given sub-quantum instant, the particle has an exact position and an exact momentum. Therefore, the theory embraces ontological determinism.

However, this determinism is masked by an epistemic indeterminism: we cannot know these exact values. Crucially, this epistemic barrier is not a technical inability (it is not a matter of lacking a powerful enough microscope). Instead, it is an unbreakable physical and statistical wall imposed by the thermodynamics of the universe and the laws of information theory.

6.2 The Physical Source: The “Smearing Out” of Momentum

In classical physics, a particle in a vacuum can be perfectly isolated. In superclassical physics, absolute vacuum does not exist. The particle is constantly buffeted by the thermal noise of the zero-point field.

Its total velocity is not just the classical forward velocity (v), but it also includes an “osmotic” or diffusive velocity (u) caused by the constant absorption and dissipation of heat. Because of this constant thermal buffeting, the particle’s classical momentum is physically “smeared out”. The underlying uncertainty in the particle’s momentum (δp) is not a magical quantum blur, but the actual root-mean-square (r.m.s.) momentum fluctuation provided by

the thermal bath:

$$\delta p = mu$$

6.3 The Fisher Length and the “Exact” Uncertainty Relation

To address the uncertainty in position without invoking waves, the authors rely on classical statistical estimation theory. In statistics, there is a concept known as the Fisher length (δx). The Fisher length is a rigorous mathematical measure that defines the absolute minimum uncertainty possible when trying to estimate the true position of a value drawn from a given probability distribution P .

When the authors mathematically relate the thermal momentum fluctuations imparted by the vacuum (δp) to the statistical Fisher length of the particle’s spatial distribution (δx), they discover a perfectly balanced, exact equation:

$$\delta x \cdot \delta p = \frac{\hbar}{2}$$

Notice that this is an exact equality ($=$), not an inequality. At the sub-quantum level, the statistical noise of the position and the thermal noise of the momentum are perfectly coupled, with Planck’s constant ($\hbar$) serving as the fundamental scaling factor for these classical thermodynamic fluctuations.

6.4 The Cramér-Rao Bound

How do we get from this sub-quantum exact equality to the famous macroscopic Heisenberg inequality ($\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$)? This is where the misconception of “technical inability” is definitively destroyed.

The transition relies on a fundamental theorem of classical statistics and information theory: the Cramér-Rao inequality. This theorem states that the variance (or error, Δx) of any macroscopic estimator or measuring device in the universe must always be greater than or equal to the optimal intrinsic variance of the underlying probability distribution (which is the Fisher length, δx).

Mathematically:

$$\Delta x \geq \delta x \quad \text{and} \quad \Delta p \geq \delta p$$

By multiplying these macroscopic measurement limits, the Heisenberg Uncertainty Principle

immediately emerges:

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

This proves that the uncertainty principle is not a reflection of clumsy human technology. The Cramér-Rao bound is an impassable mathematical law of nature. Since our macroscopic world and our instruments are entirely built upon the averaged outcomes of billions of sub-quantum collisions, any attempt to extract information from the system is inherently bottlenecked by the variance of the underlying distribution. You cannot build a machine to bypass the Cramér-Rao bound any more than you can build a machine to make $1 + 1 = 3$.

6.5 Measurement as Thermodynamic Destruction

To perfectly round out this understanding, we must ask: why can't we simply freeze the system for an instant to measure both?

In orthodox theory, looking at a particle forces the wavefunction to collapse. In the superclassical model, measurement is an irreversible thermodynamic interaction. For the particle to maintain its stable, wave-like quantum behavior, its convective velocity (v) and its diffusive velocity (u) must maintain a delicate, statistical “average orthogonality” ($\overline{v \cdot u} = 0$).

When you insert a measuring device into the setup to accurately pinpoint the particle's position ($\Delta x \rightarrow 0$), you are introducing a macroscopic obstacle that physically alters the boundary conditions (the “vacuum landscape”). This interaction forces the particle to exchange a completely stochastic, uncontrollable amount of heat with the measuring device and the zero-point field.

This thermal kick instantly breaks the average orthogonality required for stability. By extracting the positional data, you have irreversibly injected chaotic thermal momentum (Δp) into the particle.

6.6 Summary

The superclassical model completely redefines the Heisenberg Uncertainty Principle. It is not an intrinsic fuzziness of matter, but the inescapable “smearing out” of a classical particle's momentum due to the “osmotic” process of the zero-point field. We are barred from knowing the exact parameters not because we lack advanced technology, but because extracting information from a statistical thermal bath is strictly governed by the Cramér-Rao bound, and the very act

of macroscopic measurement irreversibly disrupts the thermodynamic balance of the system.

7 Conclusion

Throughout this document, we have explored how the seemingly paradoxical phenomena of quantum mechanics—such as energy quantization, multiple steady states, entanglement, and the Heisenberg Uncertainty Principle—do not necessarily require us to abandon classical logic or physical reality. By adopting the framework of Superclassical Thermodynamics, we can understand the quantum world not as a realm of abstract, probabilistic wavefunctions falling from the sky, but as an emergent macroscopic phenomenon. It is the result of continuous, deterministic thermodynamics driven by the interaction between dissipative particles and the fluctuating sub-quantum zero-point field in ordinary 3D space.

While this model successfully restores a tangible, realist ontology to physics and elegantly resolves the wave-particle duality and the measurement problem, it is not the final answer. It opens the door to new profound challenges, particularly the need for an "emergent relativity" to reconcile systemic non-locality with Einstein's Special Relativity. Ultimately, Superclassical Thermodynamics proves that quantum mechanics may not be the fundamental layer of nature, but rather a fascinating stepping stone toward a deeper understanding of the universe.